Standard Atomic Notation

atomic mass → 19

atomic number (Z#) → 9

$$^{19}_{9}\mathrm{F}$$

Z# = # of protons

= # electrons of neutral atom

mass = $p^{+} + n^{0}$

Isotopic Abundance

Recall: An *isotope* is an atom of the same element with the same number of protons and electrons, but different number of neutrons.

Not all isotopes of an element occur to the same extent in natural resources.

Ex:	*Isotope Mass*	*% Natural Abundance*
Oxygen-16	15.9949 μ	99.757
Oxygen-17	16.991 μ	0.039
Oxygen-18	17.999 μ	0.204

Recall: **μ is the Atomic Mass Unit (AMU)**

1 μ = 1.7 × 10^{-24} g

The atomic mass of an element is the average of all the isotopic masses of that element.

$^{16}_{8}O$

$$\text{Average Weight} = \frac{(\%A)(Mass.A)}{100} + \frac{(\%B)(Mass.B)}{100} + \frac{(\%C)(Mass.C)}{100} + \ldots$$

$$= \frac{(99.757)(15.9949)}{100} + \frac{(0.039)(16.991)}{100} + \frac{(0.204)(17.999)}{100}$$

$$= 15.999\mu$$

Natural potassium consists of 93.10% K-39 and 6.90% K-41. Do these relative values confirm the accepted average atomic mass for natural potassium?

$$\text{atomic mass} = \frac{(93.10)(39)}{100} + \frac{(6.90)(41)}{100}$$

$$= 39.138\,\mu$$

μ = amu = "atomic mass units"

Natural argon contains 99.60% Ar-40, 0.34% Ar-36, and 0.06% Ar-38. Do these relative values confirm the accepted average atomic mass for argon?

Radioisotopes & Half-life

Radioisotopes are isotopes that are radioactive. This means they spontaneously emit radiation in the form of particles and/or gamma rays.

The time it takes for one-half the nuclei in a radioactive sample to decay is known as its **half-life**.

$$m_t = m_i \left(\frac{1}{2}\right)^{t \div T_{1/2}}$$

m_t - final mass after time t
m_i - initial mass
t - time
$T_{1/2}$ - the half-life of the radioisotope

Ex) The half-life of cesium-137 is 30 a. What mass of Cs-137 would remain from a 12 g sample 30a? 45a? 100a?

$$m_t = m_i\left(\frac{1}{2}\right)^{t \div T_{1/2}}$$

$$m_{30} = (12)\left(\frac{1}{2}\right)^{30 \div 30}$$
$$= 12\left(\frac{1}{2}\right)^{1}$$
$$= 6\ g$$

$$m_{45} = 12\left(\frac{1}{2}\right)^{45 \div 30}$$
$$= 12\left(\frac{1}{2}\right)^{1.5}$$
$$= 4.2\ g$$

$$m_{100} = 12\left(\frac{1}{2}\right)^{100 \div 30}$$
$$= 1.19\ g$$

Read pg 17-18
Do pg 19 #1-5

+ worksheet
+ lab book for Monday!